An Ontology-Driven ESG Metric Knowledge Graph Framework for Analytics and Automated Reporting

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Chapter 4

ESG Metric Knowledge Graph

4.1 Introduction

Environmental, Social, and Governance (ESG) reporting has become a critical component of corporate disclosure, driven by increasing regulatory requirements and stakeholder demand for transparency [1]. Organisations must navigate complex relationships between reporting frameworks, disclosure categories, metrics, computational models, and underlying data sources. Regulatory frameworks such as the IFRS Sustainability Disclosure Standards (IFRS S1 and S2) [2], the Task Force on Climate-related Financial Disclosures (TCFD) [3], and the Sustainability Accounting Standards Board (SASB) [4] define what organisations should disclose. However, these frameworks often leave open how metrics are operationalised, computed, and traced back to their data origins. This gap introduces challenges in metric selection, consistency, transparency, and auditability—particularly when organisations must comply with multiple frameworks simultaneously.

Knowledge graphs have emerged as a powerful paradigm for representing complex, interconnected domain knowledge in a machine-interpretable form [5]. By encoding entities and their relationships as structured semantic networks, knowledge graphs enable systematic querying, reasoning, and integration across heterogeneous information sources [6].

These capabilities are well-suited to the challenges of ESG reporting, where traceability, transparency, and interoperability are paramount.

To address these challenges, this chapter introduces the *ESG Metric Knowledge Graph* (ESGMKG), a domain-specific knowledge graph designed to explicitly model the semantic relationships among industries, ESG reporting frameworks, categories, metrics, computational models, implementations, dataset variables, and data sources. ESGMKG provides a structured representation of ESG reporting knowledge that enables systematic navigation from high-level industry requirements down to the concrete data variables and their sources.

ESGMKG serves three key purposes:

1. **Semantic Traceability**: Establishing explicit links between reported metrics and their underlying data sources and computational models.
2. **Flexible Recomposition**: Enabling metric selection and recomposition across different reporting frameworks.
3. **Formal Foundation**: Providing a machine-interpretable foundation for querying, reasoning, and automation in ESG reporting systems.

The remainder of this chapter is organised as follows. Section 4.2 describes the ontology design methodology, including the hybrid development approach and validation strategy. Section 4.3 presents the conceptual design of ESGMKG, detailing the core entities, relationships, and modelling decisions. Section 4.4 demonstrates ESGMKG through instantiation, competency question evaluation, and example queries. Section 4.5 concludes with a discussion of limitations and future research directions.

4.2 Design Methodology

This section describes the methodology used to develop ESGMKG, including the ontology engineering approach and the validation strategy based on competency questions.

4.2.1 Ontology Engineering Approach

The development of ESGMKG follows established ontology engineering principles [7, 8]. Specifically, the ontology adopts a hybrid approach that integrates both top–down and bottom–up design strategies. This hybrid methodology is necessary because ESG reporting encompasses two distinct types of knowledge: (1) the stable conceptual structures prescribed by regulatory frameworks, and (2) the heterogeneous operational practices used by organisations and data providers.

Top–Down Design. The top–down perspective begins with the formal structures defined by ESG reporting frameworks. Frameworks such as IFRS S1 [2] and TCFD [3] prescribe high-level disclosure requirements organised into categories such as environmental risk, environmental opportunity, social risk, and governance risk. These framework-defined structures serve as conceptual anchors that ensure alignment between ESGMKG and the terminology and logic of regulatory requirements. By deriving the upper levels of the ontology from authoritative framework definitions, ESGMKG maintains consistency with established ESG reporting standards.

Bottom–Up Design. The bottom–up perspective incorporates the operational realities of ESG data workflows. In practice, ESG metrics are often the product of multi-step transformations involving estimation, aggregation, or normalisation [9]. To accurately reflect these complexities, ESGMKG models metrics alongside their computational models, implementations, dataset variables, and data sources. This bottom–up grounding ensures that the ontology captures not only what should be reported, but also how reported values

are derived from underlying data.

Hybrid Integration. By combining top–down and bottom–up perspectives, ESGMKG bridges regulatory intent and operational reality. The resulting ontology captures both the conceptual foundations of ESG reporting and the practical workflows through which ESG information is produced, validated, and consumed. This hybrid approach aligns with established recommendations for domain ontology development [8,10].

4.2.2 Validation via Competency Questions

Ontology validation is performed using competency questions (CQs), a widely adopted technique for verifying that an ontology captures the knowledge required to support its intended use cases [7,10]. Competency questions represent realistic information needs that the ontology must be able to answer. If the instantiated knowledge graph can answer all competency questions consistently and unambiguously, the ontology is considered validated for its intended purpose.

For ESGMKG, seven competency questions are defined spanning the full knowledge graph structure: identifying which reporting frameworks apply to a given industry; retrieving the categories included within a framework; discovering which metrics belong to each category; determining how a metric is calculated or measured; identifying which metrics serve as inputs for a model; finding which implementation executes a model; and tracing dataset variables to their original data sources. Table 4.6 in Section 4.4.2 presents the complete set of competency questions used to validate ESGMKG.

4.3 Conceptual Design of ESGMKG

This section presents the conceptual design of ESGMKG, including the core entities, relationships, and the rationale behind key modelling decisions.

4.3.1 Core Entities

ESGMKG comprises eight core entities that together represent the full chain of information underpinning ESG reporting. Figure 4.1 illustrates these entities and their relationships.

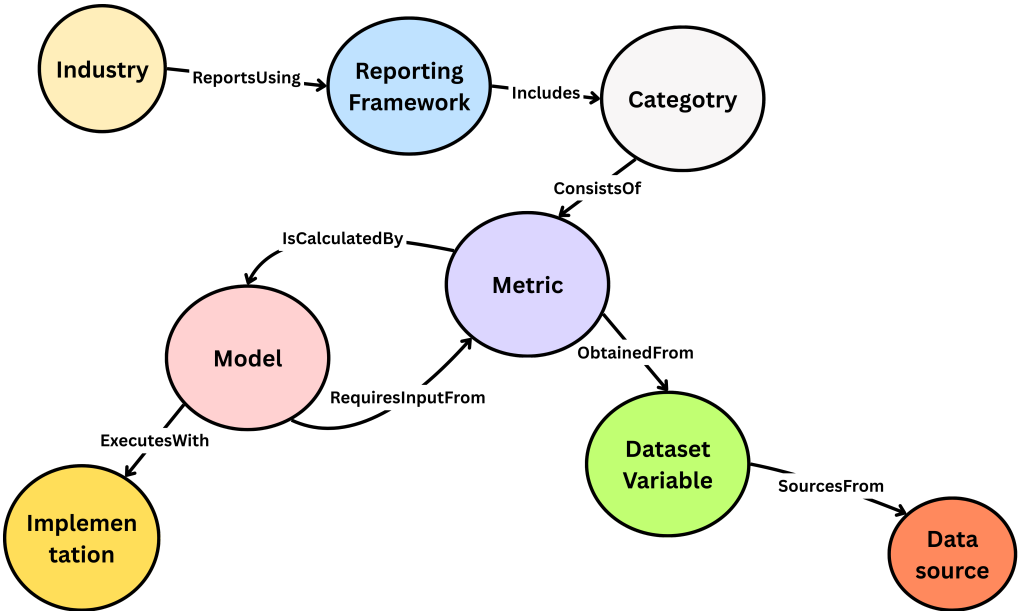

Figure 4.1: Conceptual structure of ESGMKG showing eight core entities and their semantic relationships.

Industry. An *Industry* represents the sector or domain in which an organisation operates, such as financial services, energy, or manufacturing. Industries serve as the entry point to the knowledge graph, as different industries are subject to different reporting frameworks based on regulatory requirements and stakeholder expectations.

Reporting Framework. A *Reporting Framework* represents a standardised ESG reporting standard or guideline. Examples include IFRS S1 [2], TCFD [3], and SASB [4]. Each framework defines a set of disclosure requirements organised into reporting categories.

Category. A *Category* organises ESG disclosures into thematic areas. Common categories include environmental risk, environmental opportunity, social risk, and governance risk. Categories provide a logical grouping of related metrics within a framework.

Metric. A *Metric* represents a measurable quantity used to evaluate ESG performance. Examples include carbon emissions intensity, water withdrawal, and employee turnover rate. Metrics may be derived through two distinct paths: they can be *calculated by* computational models that aggregate or transform input data, or they can be *obtained directly from* dataset variables when the metric represents a direct measurement. Metrics can also serve as inputs to other metrics, enabling the representation of composite indicators.

Model. A *Model* captures the computational logic used to derive a metric from its inputs. Models specify the mathematical operations, aggregation rules, or transformation functions applied to input data. By representing models as explicit entities, ESGMKG ensures that computational processes are transparent and auditable.

Implementation. An *Implementation* represents the executable artefact that operationalises a computational model, such as a Python script, SQL query, or web service. Separating implementations from models allows the same computational logic to be realised through different technical means, facilitating reuse and platform independence.

Dataset Variable. A *Dataset Variable* represents a raw or processed data field used in ESG computations. Examples include `CO2_Scope1`, `Electricity_Consumed`, and `Employee_Count`. Dataset variables are the atomic data elements from which metrics are derived.

Data Source. A *Data Source* identifies where dataset variables originate. Data sources may be internal enterprise systems (e.g., ERP systems, sustainability management platforms) or external ESG data providers (e.g., Eurofidai, MSCI, Refinitiv) [9].

4.3.2 Relationships

The eight core entities are connected through semantically meaningful relationships that enable navigation and traceability across the knowledge graph. Table 4.1 summarises these relationships.

Table 4.1: ESGMKG relationships connecting core entities.

Relationship	Domain	Range	Description
`ReportsUsing`	Industry	Framework	Industry reports using framework
`Includes`	Framework	Category	Framework includes categories
`ConsistsOf`	Category	Metric	Category consists of metrics
`IsCalculatedBy`	Metric	Model	Metric is calculated by model
`RequiresInputFrom`	Model	Metric / Variable	Model requires inputs
`ExecutesWith`	Model	Implementation	Model executes with implementation
`ObtainedFrom`	Metric	Variable	Metric obtained from variable
`SourcesFrom`	Variable	Data Source	Variable sources from data source

These relationships collectively enable end-to-end traceability from industry-level reporting requirements down to the specific data fields that underpin reported outcomes. Notably, metrics support two derivation paths: the `IsCalculatedBy` relationship links metrics to computational models for complex derived indicators, while the `ObtainedFrom` relationship links metrics directly to dataset variables for simple measured values. This dual-path design reflects the reality that ESG reporting includes both computed aggregations and direct measurements.

4.3.3 Schema Semantics and Traceability

The ESGMKG schema is designed around the principle of *semantic traceability* [11]. Every relationship forms part of a navigable chain that allows users to trace information from high-level reporting requirements down to raw data fields. This capability is essential for transparency and auditability in ESG reporting.

The semantics of each relationship are carefully chosen to reflect the underlying domain

meaning. For example, the relationship `IsCalculatedBy` between a metric and its model reinforces that metrics are *derived* through computation rather than directly observed. The relationship `ObtainedFrom` captures the alternative case where metrics represent direct measurements from dataset variables. The relationship `RequiresInputFrom` captures computational dependencies, while `SourcesFrom` emphasises data provenance. By grounding each relationship in a clear semantic role, ESGMKG ensures consistent interpretation across the ESG reporting lifecycle.

4.3.4 Design Rationale

Several key modelling decisions guide the design of ESGMKG:

Separation of Metrics and Dataset Variables. Metrics are conceptually distinct from dataset variables. Metrics represent the quantities that organisations report, while dataset variables represent the underlying data fields from which metrics are derived. This separation allows metric definitions to remain stable even as data sources or computational methodologies evolve [12].

Dual Derivation Paths for Metrics. Metrics can be derived through two alternative paths: calculation via models (`IsCalculatedBy`) or direct obtainment from dataset variables (`ObtainedFrom`). Both paths represent valid ways to derive a metric at the same conceptual level. This design reflects the reality that some ESG metrics require complex computation (e.g., carbon intensity per revenue) while others represent direct measurements (e.g., total water withdrawal).

Explicit Representation of Models. Computational models are represented as first-class entities rather than embedded in metric definitions or external code. This design improves transparency by making computational logic explicit, enables model reuse across

different metrics, and supports auditability by documenting how reported values are derived.

Separation of Models and Implementations. Models capture abstract computational logic, while implementations represent concrete executable artefacts. This separation follows the principle of separating concerns [13] and allows the same model to be realised through different technical implementations across different platforms or programming languages.

Recursive Metric Composition. The `RequiresInputFrom` relationship allows models to take other metrics as inputs, enabling the representation of composite indicators. This recursive structure is essential for ESG reporting, where high-level metrics (e.g., overall environmental risk score) often depend on lower-level metrics (e.g., carbon emissions, water usage).

Framework-Agnostic Design. ESGMKG is designed to be framework-agnostic, capable of representing metrics from IFRS, TCFD, SASB, GRI, or emerging frameworks without structural modification. This flexibility ensures that the knowledge graph can accommodate the evolving ESG regulatory landscape.

4.4 ESGMKG Instantiation and Evaluation

This section demonstrates ESGMKG through instantiation with a representative example, evaluates the ontology against competency questions, and illustrates querying capabilities using SPARQL.

4.4.1 Example Instantiation

To demonstrate ESGMKG in practice, we instantiate the knowledge graph with a realistic ESG reporting scenario based on the IFRS S1 framework for the financial services industry. The example focuses on environmental metrics, illustrating both *calculated metrics* derived through computational models and *direct metrics* obtained directly from dataset variables.

4.4.1.1 Metric Derivation Paths

ESGMKG supports two alternative derivation paths for metrics, both at the same conceptual level:

- **Calculated Metric Path**: A metric is derived through a computational model using the `IsCalculatedBy` relationship. The model may require other metrics or dataset variables as inputs via the `RequiresInputFrom` relationship, enabling hierarchical composition.
- **Direct Metric Path**: A metric is obtained directly from a dataset variable using the `ObtainedFrom` relationship, without requiring any computational model.

Figure 4.2 illustrates both derivation paths. The left side shows the calculated metric path with hierarchical composition, where an aggregate metric depends on component metrics, which in turn depend on dataset variables. The right side shows the direct metric path where a metric is obtained directly from a dataset variable.

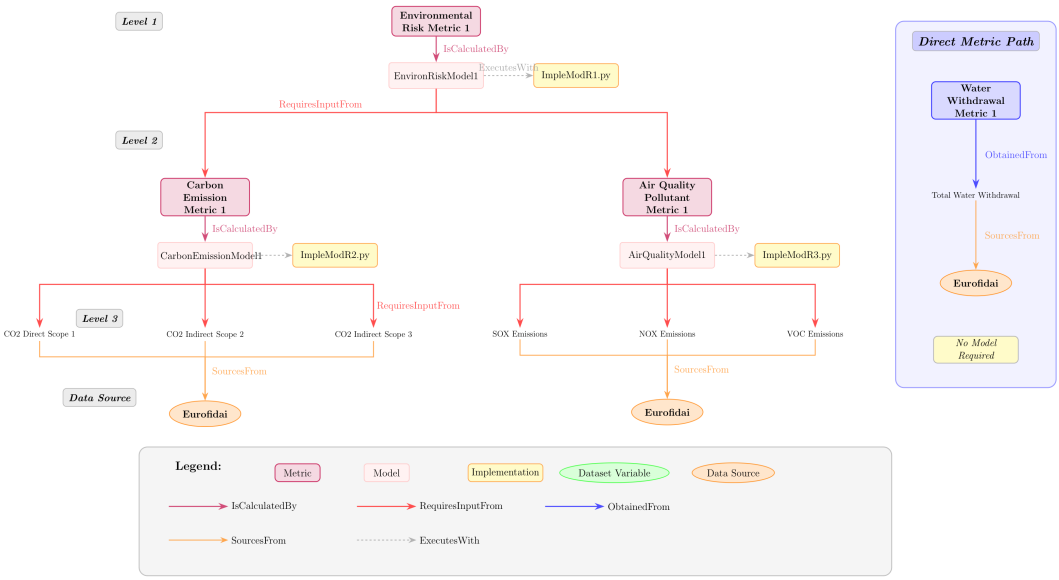

Figure 4.2: Metric derivation paths in ESGMKG. Left: Calculated metric path showing hierarchical composition where EnvironmentalRiskMetric1 (Level 1) is calculated from two component metrics (Level 2), each derived from dataset variables (Level 3). Right: Direct metric path where WaterWithdrawalMetric1 is obtained directly from a dataset variable without computation. Both paths represent alternative ways to derive metrics at the same conceptual level.

4.4.1.2 Calculated Metric Path: Hierarchical Composition

The calculated metric path demonstrates hierarchical composition with three levels:

- **Level 1 – Aggregate Metric**: *EnvironmentalRiskMetric1* is calculated by *EnvironRiskModel1*, which requires two component metrics as inputs.
- **Level 2 – Component Metrics**: *CarbonEmissionMetric1* is calculated by *CarbonEmissionModel1* from three GHG Protocol scope variables. *AirQualityPollutantMetric1* is calculated by *AirQualityModel1* from three pollutant variables.
- **Level 3 – Dataset Variables**: The raw data fields (CO2DirectScope1, CO2IndirectScope2, CO2IndirectScope3, SOXEmissions, NOXEmissions, VOCEmissions) that serve as inputs to the component metric models.

4.4.1.3 Direct Metric Path

The direct metric path demonstrates the alternative derivation approach: *WaterWithdrawalMetric1* is obtained directly from the *TotalWaterWithdrawal* dataset variable via the `ObtainedFrom` relationship. No computational model is required for this path.

4.4.1.4 Instantiation Tables

The following tables present the complete ESGMKG instantiation.

Table 4.2 shows the reporting path from industry through framework and category to metrics.

Table 4.2: ESGMKG instantiation: Reporting path from industry to metrics.

Step	Source Entity	Relationship	Target Entity
1	Industry: Financial Services	ReportsUsing	Framework: IFRS S1
2	Framework: IFRS S1	Includes	Category: Environmental Risk
3	Category: Environmental Risk	ConsistsOf	Metric: EnvironmentalRiskMetric1
4	Category: Environmental Risk	ConsistsOf	Metric: WaterWithdrawalMetric1

Table 4.3 shows the calculated metric path with hierarchical composition.

Table 4.3: ESGMKG instantiation: Calculated metric path with hierarchical composition.

Step	Source Entity	Relationship	Target Entity
Level 1: Aggregate Metric			
5	Metric: EnvironmentalRiskMetric1	IsCalculatedBy	Model: EnvironRiskModel1

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Table 4.3 – *Continued from previous page*

Step	Source Entity	Relationship	Target Entity
6	Model: EnvironRiskModel1	ExecutesWith	Impl: ImpleModR1.py
7	Model: EnvironRiskModel1	RequiresInputFrom	Metric: CarbonEmissionMetric1
8	Model: EnvironRiskModel1	RequiresInputFrom	Metric: AirQualityPollutantMetric1
Level 2: Carbon Emission Component Metric			
9	Metric: CarbonEmissionMetric1	IsCalculatedBy	Model: CarbonEmissionModel1
10	Model: CarbonEmissionModel1	ExecutesWith	Impl: ImpleModR2.py
11	Model: CarbonEmissionModel1	RequiresInputFrom	Variable: CO2DirectScope1
12	Model: CarbonEmissionModel1	RequiresInputFrom	Variable: CO2IndirectScope2
13	Model: CarbonEmissionModel1	RequiresInputFrom	Variable: CO2IndirectScope3
Level 2: Air Quality Component Metric			
14	Metric: AirQualityPollutantMetric1	IsCalculatedBy	Model: AirQualityModel1
15	Model: AirQualityModel1	ExecutesWith	Impl: ImpleModR3.py
16	Model: AirQualityModel1	RequiresInputFrom	Variable: SOXEmissions

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Table 4.3 – *Continued from previous page*

Step	Source Entity	Relationship	Target Entity
17	Model: AirQuality-Model1	RequiresInputFrom	Variable: NOXEmissions
18	Model: AirQuality-Model1	RequiresInputFrom	Variable: VOCEmissions

Table 4.4 shows the direct metric path.

Table 4.4: ESGMKG instantiation: Direct metric path (no computational model required).

Step	Source Entity	Relationship	Target Entity
19	Metric: WaterWithdrawal-Metric1	ObtainedFrom	Variable: TotalWaterWithdrawal

Table 4.5 shows the data provenance path.

Table 4.5: ESGMKG instantiation: Data provenance linking variables to data source.

Step	Source Entity	Relationship	Target Entity
20	Variable: CO2DirectScope1	SourcesFrom	DataSource: Eurofidai
21	Variable: CO2IndirectScope2	SourcesFrom	DataSource: Eurofidai
22	Variable: CO2IndirectScope3	SourcesFrom	DataSource: Eurofidai
23	Variable: SOXEmissions	SourcesFrom	DataSource: Eurofidai
24	Variable: NOXEmissions	SourcesFrom	DataSource: Eurofidai
25	Variable: VOCEmissions	SourcesFrom	DataSource: Eurofidai
26	Variable: TotalWaterWithdrawal	SourcesFrom	DataSource: Eurofidai

4.4.1.5 Summary of Instantiation

The instantiation demonstrates how ESGMKG captures the full complexity of ESG metric derivation:

- **Dual derivation paths**: Calculated metrics use `IsCalculatedBy` while direct metrics use `ObtainedFrom`, representing alternative approaches at the same conceptual level.
- **Hierarchical composition**: Within the calculated path, EnvironmentalRiskMetric1 aggregates CarbonEmissionMetric1 and AirQualityPollutantMetric1, which depend on dataset variables.
- **Multiple inputs per model**: EnvironRiskModel1 requires 2 metrics; CarbonEmissionModel1 and AirQualityModel1 each require 3 variables.
- **Data provenance**: All variables trace back to Eurofidai as the data source.
- **Complete coverage**: All 8 entity types and all 8 relationships are demonstrated.

4.4.2 Competency Question Evaluation

The instantiated ESGMKG is validated using competency questions that test its ability to support essential ESG reporting queries. Table 4.6 presents the seven competency questions and their expected outputs.

Table 4.6: Competency questions for ESGMKG validation.

CQ	Question	Expected Output
CQ1	Which framework applies to a given industry?	Reporting Framework
CQ2	What categories are included within that framework?	Category
CQ3	Which metrics belong to each category?	Metric
CQ4	How is a metric calculated or measured?	Model or Dataset Variable
CQ5	Which metrics are inputs for a model?	Metric
CQ6	Which implementation executes a model?	Implementation
CQ7	What is the original data source for a variable?	Data Source

All seven competency questions can be answered by the instantiated knowledge graph. For **CQ1**, querying which framework applies to the financial services industry returns

IFRS S1 (Table 4.2, Step 1). For **CQ3**, querying which metrics belong to the Environmental Risk category returns both EnvironmentalRiskMetric1 and WaterWithdrawalMetric1 (Steps 3–4). For **CQ4**, querying how EnvironmentalRiskMetric1 is derived returns EnvironRiskModel1 via `IsCalculatedBy` (Table 4.3, Step 5), while querying how WaterWithdrawalMetric1 is derived returns the TotalWaterWithdrawal dataset variable via `ObtainedFrom` (Table 4.4, Step 19)— demonstrating both derivation paths. For **CQ5**, querying which metrics are inputs for EnvironRiskModel1 returns CarbonEmissionMetric1 and AirQualityPollutantMetric1 (Steps 7–8). For **CQ7**, querying the original data source for CO2DirectScope1 returns Eurofidai (Table 4.5, Step 20).

4.4.3 Usage Scenarios

ESGMKG supports diverse usage scenarios relevant to ESG reporting practice. In **framework selection**, an organisation identifies applicable reporting frameworks by querying ESGMKG with their industry; for example, a financial services firm can discover that IFRS S1 and TCFD are relevant frameworks for their sector. In **metric discovery**, a sustainability officer explores reporting categories to identify available metrics and understand their derivation paths, revealing not only what metrics exist but also whether they are calculated or directly obtained. In **audit trail** scenarios, an auditor traces a reported metric back through its derivation path to the underlying dataset variables and data sources, supporting assurance and regulatory compliance. In **model reuse** scenarios, a data engineer identifies computational models that can be reused across different metrics or frameworks, along with their existing implementations, reducing development effort and ensuring consistency.

4.4.4 RDF Representation and SPARQL Querying

ESGMKG is implemented using the Resource Description Framework (RDF) [14] and the Web Ontology Language (OWL) [15], ensuring interoperability with semantic web technologies. Each entity and relationship is expressed as RDF triples that can be queried

using SPARQL [16]. Table 4.7 illustrates example RDF triples representing key entities and relationships from the instantiation.

Table 4.7: Example RDF triples in ESGMKG.

Subject	**Predicate**	**Object**
esgmkg:FinancialServices	rdf:type	esgmkg:Industry
esgmkg:FinancialServices	esgmkg:ReportsUsing	esgmkg:IFRS_S1
esgmkg:IFRS_S1	esgmkg:Includes	esgmkg:EnvRisk
esgmkg:EnvRisk	esgmkg:ConsistsOf	esgmkg:ERM1
esgmkg:ERM1	esgmkg:IsCalculatedBy	esgmkg:EnvRiskModel1
esgmkg:EnvRiskModel1	esgmkg:RequiresInputFrom	esgmkg:CEM1
esgmkg:EnvRiskModel1	esgmkg:ExecutesWith	esgmkg:ImpleModR1
esgmkg:WaterMetric1	esgmkg:ObtainedFrom	esgmkg:TotalWater
esgmkg:CO2Scope1	esgmkg:SourcesFrom	esgmkg:Eurofidai

SPARQL queries enable flexible retrieval of information from ESGMKG. Algorithm 1 shows a query that retrieves all metrics in a given category along with their derivation method, demonstrating how to distinguish between calculated and direct metrics.

Algorithm 1 SPARQL query to retrieve metrics and their derivation methods.

```
PREFIX esgmkg: <http://example.org/esgmkg#>

SELECT ?metric ?model ?variable
WHERE {
    esgmkg:EnvironmentalRisk esgmkg:ConsistsOf ?metric .
    OPTIONAL { ?metric esgmkg:IsCalculatedBy ?model . }          ▷ Calculated
    OPTIONAL { ?metric esgmkg:ObtainedFrom ?variable . }         ▷ Direct
}
```

Algorithm 2 presents a query that traces a calculated metric through its full hierarchy to the underlying data sources.

Algorithm 2 SPARQL query to trace metric hierarchy to data sources.

```
PREFIX esgmkg: <http://example.org/esgmkg#>

SELECT ?topMetric ?inputMetric ?variable ?dataSource
WHERE {
    ?topMetric esgmkg:IsCalculatedBy ?topModel .                    ▷ Level 1
    ?topModel esgmkg:RequiresInputFrom ?inputMetric .
    ?inputMetric esgmkg:IsCalculatedBy ?inputModel .                ▷ Level 2
    ?inputModel esgmkg:RequiresInputFrom ?variable .
    ?variable esgmkg:SourcesFrom ?dataSource .                      ▷ Level 3
}
```

These queries illustrate how ESGMKG enables systematic navigation through both metric derivation paths, from high-level metrics down to underlying data sources.

4.5 Discussion and Conclusion

4.5.1 Summary of Contributions

This chapter presented ESGMKG, a domain-specific knowledge graph for ESG metric management. The primary contribution is a **conceptual model** comprising eight core entities—Industry, Reporting Framework, Category, Metric, Model, Implementation, Dataset Variable, and Data Source—connected through eight semantic relationships that enable end-to-end traceability. A key design feature is the **dual derivation paths** for metrics: calculated metrics are linked to computational models via `IsCalculatedBy`, while direct metrics are linked to dataset variables via `ObtainedFrom`. Both paths represent alternative approaches at the same conceptual level. The **hierarchical metric composition** capability, demonstrated within the calculated metric path, shows how aggregate metrics can depend on component metrics, which in turn depend on raw dataset variables. The

ontology was developed using a **hybrid design methodology** that combines top–down structures derived from regulatory frameworks with bottom–up grounding in operational ESG data workflows. The chapter **validated** the ontology through seven competency questions representing realistic ESG reporting information needs.

4.5.2 Limitations

Several limitations of the current work should be acknowledged. The **scope of instantiation** focuses on environmental risk metrics for demonstration purposes; comprehensive deployment would require extensive population with metrics from multiple categories (social, governance) and frameworks across different industries. The schema does not explicitly model **temporal dynamics** such as framework versioning, metric evolution, or historical changes to computational models; incorporating temporal semantics [17] would enable tracking of how ESG reporting requirements evolve over time. While ESGMKG traces metrics to data sources, it does not capture **data quality metadata** such as completeness, accuracy, or timeliness; extending the ontology with data quality dimensions [18] would enhance its utility for assurance purposes. Finally, the current approach relies on **manual population** of the knowledge graph; future work could explore automated extraction of ESG knowledge from regulatory documents and corporate reports using natural language processing techniques [19].

4.5.3 Future Research Directions

Future research directions include **extended framework coverage** to incorporate additional frameworks such as GRI, CDP, and TNFD, along with social and governance metrics; **integration with large language models** to enable natural language querying and automated knowledge extraction from unstructured ESG documents [20]; **reasoning and inference** capabilities using OWL reasoning to infer implicit relationships, detect inconsistencies, and recommend appropriate metrics based on industry context; and **enterprise integration** patterns for connecting ESGMKG with enterprise sustainability

management systems and ESG reporting platforms.

4.5.4 Conclusion

ESGMKG addresses the critical need for structured, machine-interpretable representations of ESG reporting knowledge. By explicitly modelling the relationships between frameworks, metrics, models, and data sources—and by supporting both calculated and direct metric derivation paths with hierarchical composition—ESGMKG enables transparency, traceability, and interoperability in ESG reporting. The knowledge graph provides a foundation for building intelligent ESG reporting systems that can navigate the complexity of multi-framework compliance while maintaining auditability and consistency.

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